1. Introduction: Problem and Motivation

1.1 Problem Statement

Many adults lack fundamental financial management skills, a problem that often originates in childhood due to limited exposure to financial education at an early age. In many education systems, financial literacy is minimally covered or entirely absent, resulting in children developing poor financial habits that can persist into adulthood and negatively affect their long-term financial well-being.

Although several applications attempt to address financial literacy for children, many existing solutions prioritize parental control or gamified rewards rather than meaningful, child-centered learning. Applications such as BusyKid and Goalino emphasize chore completion and monetary incentives, while providing limited direct instruction on financial concepts for children. Other tools, such as Kidonomic, attempt to teach financial literacy through text-based, chatbot-driven interaction, which may not be well suited for younger users with developing literacy and typing skills. As a result, there remains a lack of developmentally appropriate financial literacy applications that effectively teach core financial concepts directly to children.

1.2 Target Audience

The primary target audience for this application is children aged 9 to 11, corresponding to students in Grades 4 to 6. At this stage, learners are developing foundational mathematical, literacy, and decision-making skills and are capable of understanding basic financial concepts. However, these concepts are often not formally introduced in school curricula.

Children in this age group engage most effectively with short, interactive content rather than text-heavy instruction. As a result, the application is designed to support autonomous, self-paced learning through visually driven and interactive lessons, while maintaining sufficient structure to guide learners and prevent cognitive overload.

1.3 Motivation

The motivation behind this application is to introduce financial literacy at an early age to support long-term financial well-being. Early exposure allows foundational concepts to be reinforced over time, making them easier to apply later in life. By providing an engaging, age-appropriate learning experience for children aged 9 to 11, this application aims to help learners develop essential financial skills in a way that is accessible, meaningful, and motivating.

2. Background and Related Work

2.1 Learning and Educational Foundations

The design of this application is grounded in established educational principles that emphasize clear learning goals, structured content delivery, and age-appropriate instruction. While learning outcomes play an important role in guiding educational design and evaluation, they are not presented directly to the

primary users of the application. For children aged 9 to 11, explicitly stating learning outcomes within the interface may introduce unnecessary text and increase cognitive load. Instead, learning outcomes are intended for parents and educators, who may wish to understand the educational objectives and progression of the application. These outcomes are therefore made available externally, allowing the application itself to remain visually simple and learner-focused [1].

Rather than explicitly displaying learning outcomes to learners, the application evaluates achievement indirectly through embedded assessments and interactive tasks. After completing a lesson, students are encouraged to attempt short assessments that reinforce key concepts. Successful completion provides immediate feedback, allowing learners to recognize their understanding, while unsuccessful attempts prompt opportunities to revisit the lesson. This approach aligns with formative assessment practices, which emphasize learning through feedback rather than performance pressure.

To further support effective learning, the application adopts a microlearning approach, delivering content in short, focused lessons rather than lengthy instructional sessions. Microlearning supports engagement and retention by presenting information in manageable segments, which is particularly beneficial for younger learners with limited attention spans. Financial literacy concepts are therefore introduced incrementally to reduce cognitive load and support long-term understanding [2].

In addition, lessons are designed to be interactive rather than passive, allowing learners to actively engage with financial concepts instead of simply consuming information. Interactive learning objects encourage exploration, decision-making, and reflection, contributing to stronger conceptual understanding.

The application also incorporates principles of autonomous learning, allowing students to choose which lessons or topics they wish to explore. Autonomy has been shown to increase engagement and intrinsic motivation when learners feel a sense of control over their educational experience. However, autonomy within this application is intentionally balanced with structure: lessons are organized into units and guided pathways to prevent learners from becoming overwhelmed or disengaged [3].

Finally, the application is designed to encourage learning through mistakes. Rather than penalizing errors, the system allows students to make mistakes in a low-risk environment and provides constructive feedback to guide improvement. Such environments support resilience, confidence, and deeper learning by reducing fear of failure and encouraging persistence [4].

2.2 UX & Learner Experience Design

The visual and interaction design of the application is informed by both User Experience (UX) and Learner Experience Design (LXD) principles. While UX focuses on usability and efficiency, LXD extends beyond interface design to consider how users learn, engage, and progress toward educational goals. In educational applications, design decisions must support learning outcomes rather than focusing solely on visual appeal or interaction speed [5].

For younger learners, visual communication plays a particularly important role. Research indicates that images are processed more quickly than text, while reading requires greater cognitive effort due to

sequential decoding. This distinction is especially relevant for children who may not yet have advanced literacy skills. To ensure learners focus on understanding financial concepts rather than decoding language, the application prioritizes visual elements, icons, and interactive components, while minimizing dense text [6].

The interface is also designed to be visually engaging and clear, drawing from common design practices in children's educational materials. Visual consistency, clear affordances, and supportive use of color are used to guide attention and reduce unnecessary cognitive effort, ensuring that interface elements support learning rather than distract from it.

Through the application of LXD principles, user flows, lesson structures, and feedback mechanisms are intentionally aligned with how children learn, ensuring that usability decisions directly support educational effectiveness.

2.3 Gamification & Motivation

Gamification plays a central role in sustaining engagement within the application. When learning activities are framed as game-like experiences rather than traditional instruction, children are more likely to participate willingly and remain motivated. Gamification incorporates elements such as points, progression systems, and avatars to encourage continued interaction while preserving educational intent [7].

In this application, gamification functions as a supportive mechanism rather than a substitute for learning. Points and badges are awarded only after meaningful engagement with educational content and successful completion of assessments, ensuring that rewards reinforce learning rather than encourage superficial interaction.

A key gamification element is the inclusion of a personalized in-game avatar or pet. Children often form emotional connections with virtual characters, which can support long-term engagement and a sense of responsibility. Allowing learners to customize and care for their avatar using in-game currency earned through learning activities strengthens the connection between effort and reward, while maintaining focus on educational goals.

Competitive features such as leaderboards are intentionally excluded. While competition can motivate some learners, social comparison at a young age may lead to discouragement or disengagement for others. Instead, the application emphasizes personal progress and mastery, creating a more inclusive and supportive motivational environment.

Additionally, the application introduces simplified investment-related mechanics, such as an in-game stock market, to familiarize learners with basic financial decision-making in a controlled, low-risk environment. These mechanics are designed to reinforce foundational concepts rather than simulate real-world financial risk.

2.4 Universal Design for Learning (UDL)

To support accessibility and inclusivity, the application follows Universal Design for Learning (UDL) principles. UDL emphasizes providing multiple means of engagement, representation, and expression to accommodate diverse learners. By incorporating visual content, interactive tasks, flexible pacing, and learner choice, the application supports a wide range of learning preferences and abilities.

These design choices ensure that the application remains accessible to learners with varying cognitive strengths, attention levels, and learning needs.

2.5 Learning Analytics

The application integrates basic learning analytics to monitor progress and support instructional improvement. Metrics such as lesson completion, assessment attempts, and concept mastery are tracked at a high level to evaluate learning effectiveness. These analytics are used to refine lesson design and provide insight to parents or educators, while maintaining ethical and age-appropriate data practices.

3. App Design

3.1 Overall Learning Objective and Learning Outcomes

The overall learning objective of this application is to help students aged 9–11 develop foundational financial literacy skills through interactive, scenario-based activities that support informed decision-making, goal setting, and reflection on financial choices.

After engaging with the application, learners will be able to:

- Identify and classify income and expenses in simulated financial scenarios.
- Distinguish between needs and wants and apply this understanding when making financial decisions.
- Set and monitor short-term savings goals within the application.
- Analyze the outcomes of financial decisions presented in interactive scenarios.
- Reflect on financial choices and adjust decisions based on feedback and repeated attempts.

These learning outcomes guide the design of lessons, assessments, and interactive features throughout the application.

3.2 User Flow and Storyboard

This storyboard presents two primary use cases that illustrate the core user flow of the application: user login and dashboard access, and lesson navigation and engagement.

In the first use case, students launch the application and log in by entering basic information such as their name and grade level. Upon successful login, users are directed to the main dashboard, which provides an overview of learning progress, available navigation tabs, and current in-game currency. From this dashboard, learners can access lessons, assessments, investment simulations, and avatar customization features.

The second use case focuses on lesson engagement. From the dashboard, users navigate to the lessons page, where grade-appropriate units are displayed. Learners select a lesson and begin an interactive learning experience designed to introduce and reinforce financial concepts. After completing a lesson, users may engage in assessments or activities that provide feedback and allow them to reflect on their decisions.

This user flow supports the learning outcomes outlined in Section 3.1 by guiding learners through structured, interactive lessons that promote understanding, decision-making, and learning through feedback.

Figures 1 and 2 illustrate the login flow and lesson navigation storyboard described above.

Figure 1:

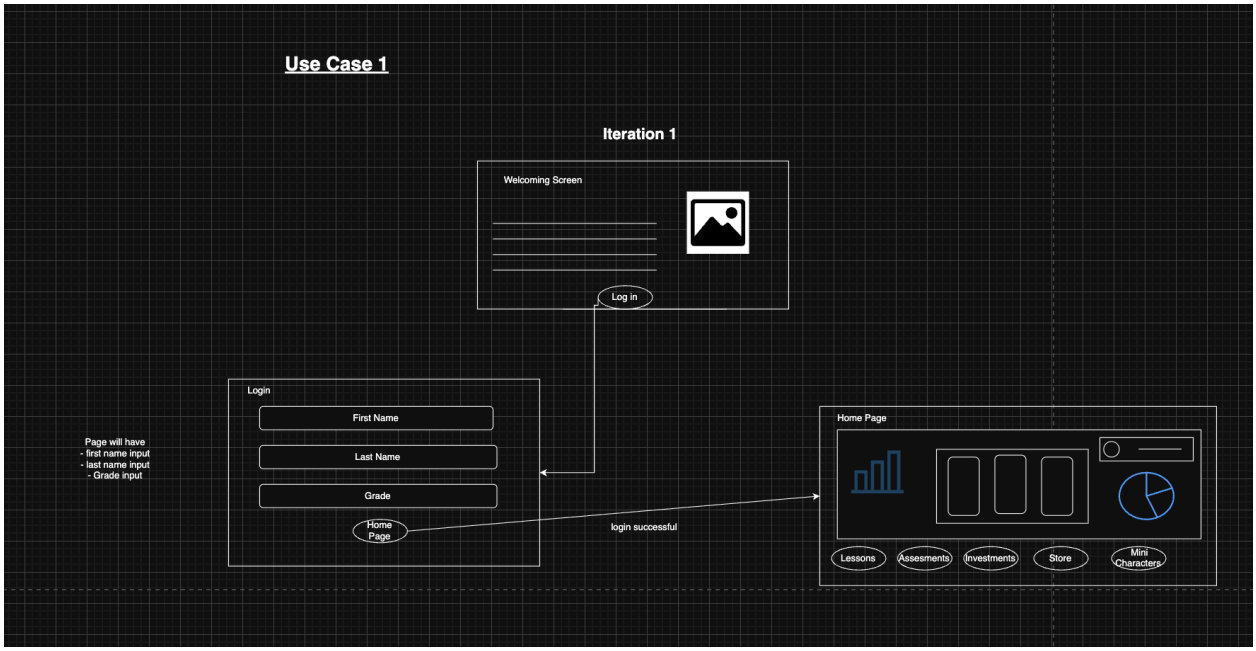
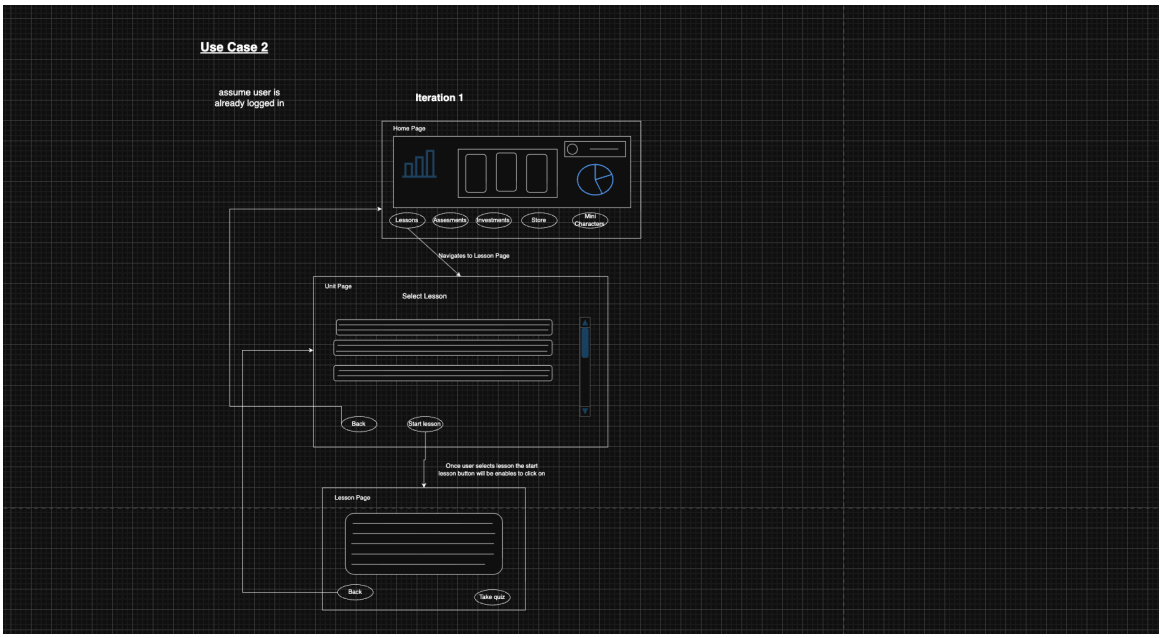


Figure 2:



3.3 Sample Learning Objects

Learning Object 1: Needs vs. Wants Scenario

Learning Objective Supported: Distinguish between needs and wants

Description: Learners are presented with a series of purchasing scenarios and must classify each item as either a need or a want.

Assessment and Feedback: Immediate feedback explains each classification and allows learners to retry, supporting reflection and learning through mistakes.

Learning Object 2: Credit Card Basics

Learning Objective Supported: Understand what credit cards are and when they should be used

Description: Learners answer short, age-appropriate questions and true-or-false statements related to credit card usage and responsibility.

Assessment and Feedback: Immediate feedback clarifies misconceptions and allows learners to revise their answers.

Learning Object 3: Price Awareness Activity

Learning Objective Supported: Evaluate whether prices are reasonable in purchasing scenarios

Description: Learners are shown everyday items with different price points and must decide whether each item is fairly priced.

Assessment and Feedback: Feedback explains the reasoning behind each outcome, allowing learners to reflect and adjust their decisions.

3.4 Gamification and Feedback Integration

Gamification is used within the application to support motivation and sustained engagement while maintaining a clear focus on learning. Learners earn points by completing lessons, activities, and assessments that demonstrate understanding of financial concepts. These points function as in-game currency and can be used to customize a personal avatar or pet, reinforcing the connection between learning effort and reward.

Feedback plays a central role in the learning experience. Rather than penalizing mistakes, the application provides immediate, supportive feedback that explains why a choice is correct or incorrect. Learners are encouraged to retry activities and reflect on their decisions, promoting learning through mistakes and reinforcing key concepts.

Competitive elements such as leaderboards are intentionally excluded. Instead, the application emphasizes personal progress and mastery, creating an inclusive and low-pressure learning environment appropriate for young learners.

3.5 Interface Design and Prototype

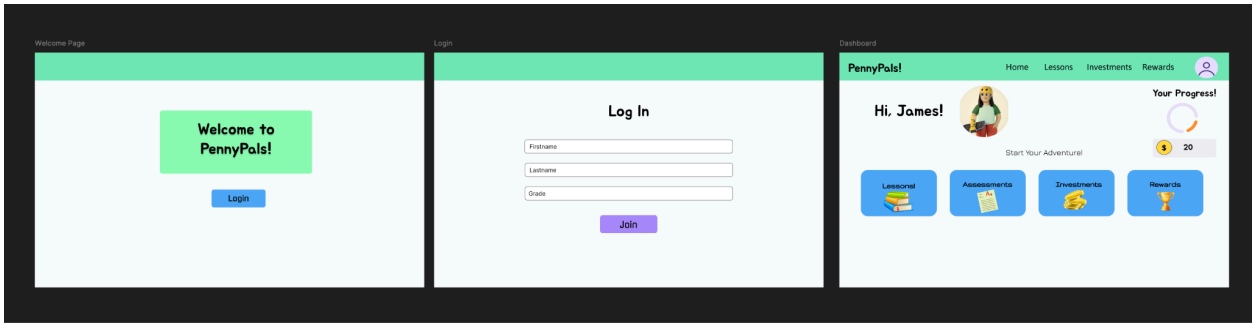
A low-fidelity interactive prototype of the application has been developed using Figma to demonstrate the core user flow and interface structure. The prototype includes key screens such as the welcome page, login screen, dashboard, lesson selection, and sample lesson interactions.

The interface design prioritizes clarity and ease of use for learners aged 9–11. Large buttons, simple layouts, and visual icons are used to support intuitive navigation and reduce cognitive load. Consistent color usage and repeated layout patterns across screens help learners recognize interface elements and move through the application with minimal confusion.

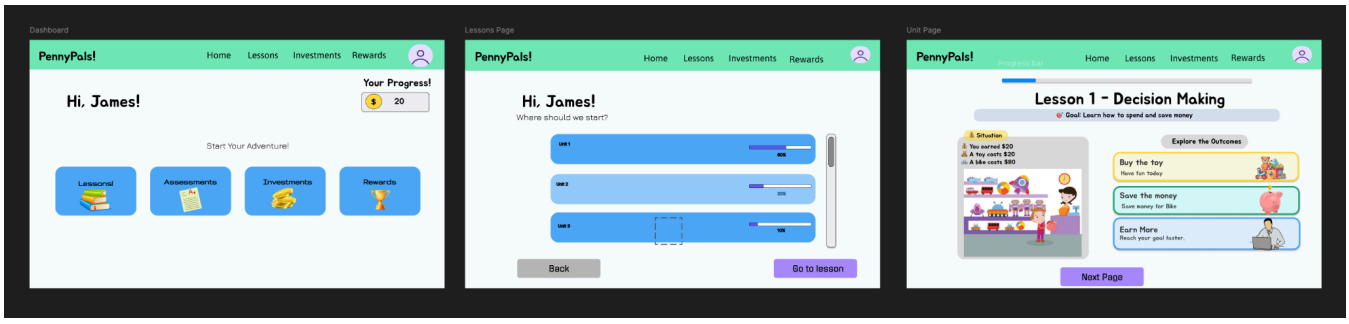
The prototype visually supports the user flows and learning interactions described in Sections 3.2 and 3.3, providing a clear representation of how learners engage with lessons, assessments, and feedback within the application.

Figma Prototype Link:
https://www.figma.com/design/a1LQrMQWEglB0j6VjNv12q/Education-App-Lab-4?node-id=0-1&t=33ZtO81lixwvM2yE-1

Use Case 1:



Use Case 2:



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5. AI Use

Chatgpt was used to rewrite the writing for grammar and better clarification. All paragraphs were typed and grammar checked and reworded for clarity purposes.